

Network Security Hardening for a Residential Apartment: defending against network and signal vector attacks

1 Overview

This design document proposes a layered security architecture for a **small residential network** (e.g., an apartment) to mitigate attack vectors like those described in training material for industrial generator controllers and in research on router exploitation and ultrasonic/AI-assisted attacks. While the training videos discussed Deep Sea Electronics modules, the attacker's knowledge of **Modbus**, **SNMP**, and **CAN** protocols can be extrapolated to general network exploitation. Additionally, attackers could leverage **MikroTik or similar routers** and **AI/ultrasonic technologies** to compromise consumer networks.

Key reference points include:

- **Exposed services and weak credentials** – Many embedded devices ship with HTTP interfaces and default passwords; DSE controllers in the videos allow up to five concurrent HTTP sessions with default credentials [8†L120-L131] . The same misconfigurations are common in consumer IoT devices.
- **MikroTik router exploitation** – A botnet of ~13 000 MikroTik routers was hijacked via old vulnerabilities and misconfigured services; attackers enabled **SOCKS proxies** and used the routers to relay malware and malspam ¹ . Unpatched MikroTik firmware such as **CVE-2025-10948** (buffer overflow in RouterOS REST API) can allow remote code execution ² .
- **Ultrasonic/AI attacks** – Research shows that inaudible voice injections over ultrasonic channels can trigger commands with success rates above **89.8 %** ³ , and that defensive perturbations can block these attacks ⁴ . Malicious actors might adapt such techniques to confuse voice assistants or sensors.

This document adapts these insights to a home network environment, recommending technical measures, architectural choices and operational practices to harden the network security software and infrastructure.

2 Assumptions and environment

- The user lives in an **apartment** with no industrial generators; however, their network may contain consumer routers (e.g., MikroTik, TP-Link), Wi-Fi access points, smart devices (voice assistants, IoT cameras) and work computers.

- The adversary is **technically capable**, possibly with access to government or contractor resources, and may try to exploit home networks via remote code execution, phishing/malspam through compromised routers, or signal-based attacks (ultrasonic or RF).
- The user wants to **harden network security software** and overall network architecture to resist these vectors while maintaining usability.

3 Threat vectors

3.1 Router and network infrastructure

1. **Unpatched firmware / RCE vulnerabilities** – Vulnerabilities like **CVE-2025-10948** in MikroTik RouterOS allow remote attackers to overflow buffers via the REST API ². Botnets have exploited older MikroTik vulnerabilities to deploy SOCKS proxies and send malspam ¹.
2. **Default credentials and open services** – Many consumer routers ship with default admin credentials and expose management services (HTTP, WinBox, Telnet). The DSE training emphasised how controllers rely on default **Admin/Password1234** credentials **[5+L83-L85]**; similar defaults exist on consumer devices.
3. **SPF/DNS misconfigurations** – The malspam botnet exploited domains whose SPF records allowed all senders (**+all**) ⁵, enabling mail spoofing and phishing.
4. **Network segmentation and lateral movement** – Without VLANs or guest networks, an attacker who compromises any IoT device can pivot to computers or smart assistants.

3.2 Endpoint and application layer

1. **Weak IoT devices** – Voice assistants, smart TVs, and cameras often lack patch management or encryption. Attackers could abuse them to listen, pivot, or run command-and-control.
2. **Inaudible command injection** – Ultrasound research (SUAD) shows adversaries can inject voice commands via solid-channel ultrasonic signals with high success rates ³. Although the research is academic, consumer devices may be unprotected.
3. **Phishing / malspam** – As shown in the MikroTik botnet case, compromised routers can send malicious email that appears to come from legitimate domains ⁵.

3.3 Physical and wireless signals

1. **RF interference and eavesdropping** – Attackers could use SDR tools to monitor Wi-Fi or attempt deauthentication attacks. Unshielded cables may radiate signals that can be captured.
2. **Ultrasonic or electromagnetic injection** – Attackers could place hidden speakers near windows or walls to inject ultrasonic commands. They could also inject EMI into power lines or Ethernet cables to corrupt data (a vector considered in DSE training when discussing proper cable shielding).

4 Security architecture

4.1 Segmented home network

1. **Core router/firewall** – Use a **hardened firewall appliance** (e.g., pfSense/OPNsense on a dedicated mini-PC) instead of consumer ISP-supplied routers. Benefits:

2. Open-source firmware with timely security patches.
3. Support for VLAN segmentation, advanced firewall rules and intrusion detection (IDS/IPS).
4. Transparent bridging for ISP modem (bridge mode), reducing attack surface.
5. **Wi-Fi access points** – Deploy separate **Wi-Fi networks**:
6. **Trusted LAN SSID** for laptops/desktops; WPA3-Personal encryption and strong passphrases.
7. **IoT VLAN/SSID** for smart devices; restrict outbound traffic to only required services (e.g., NTP, vendor cloud). Block east-west traffic among IoT devices.
8. **Guest VLAN/SSID** for visitors; isolate from other networks.
9. **Router management security**:
10. **Disable remote management**; restrict administration (HTTPS/SSH) to wired LAN or a VPN.
11. **Change default credentials** and enforce strong passwords; use two-factor authentication where supported.
12. **Restrict service ports** (disable Telnet, WinBox, API, SMB) on MikroTik or similar devices. Only enable SSH and HTTPS management.
13. **Patch firmware promptly**; subscribe to vendor security advisories. MikroTik recommends upgrading to RouterOS 7.21 or later to mitigate CVE-2025-10948 ².
14. **Disable unnecessary features** like SOCKS proxies and DNS forwarding on the router to prevent abuse ⁶.
15. **DNS and email security**:
16. Host personal domains with reputable providers and configure **restrictive SPF records** (avoid `+all`); use **DKIM** and **DMARC** to prevent spoofing ⁵.
17. Use **encrypted DNS** (DNS over HTTPS/TLS) to prevent DNS hijacking and monitoring.

4.2 Endpoint hardening

1. **Operating system patching** – Enable automatic security updates on all desktops, laptops, and mobile devices. Use a modern OS with long-term support.
2. **Endpoint firewall & EDR** – Deploy endpoint firewall rules and an **endpoint detection and response (EDR)** or antivirus solution. Monitor for unusual outbound connections or port scans.
3. **Least-privilege accounts** – Use non-administrator accounts for everyday tasks. Enable device encryption (BitLocker/FileVault).
4. **Secure IoT devices**:
5. Choose devices with a security track record; update firmware regularly.
6. Change default passwords and disable UPnP.
7. For voice assistants, review privacy settings; disable voice purchasing or call capabilities; configure **wake-word confirmation** and audible responses.
8. If concerned about ultrasonic attacks, **physically disconnect or cover microphones** when not in use or use devices with physical mute switches. Consider ultrasonic jammers or low-pass filters for

critical sensors; SUAD research suggests that ultrasonic perturbation signals can block inaudible attacks ⁴ .

4.3 Network monitoring and intrusion detection

1. **IDS/IPS** – Use Suricata or Snort on the firewall to inspect traffic for signatures of known malware and attacks. Configure rulesets for **Modbus**, **SNMP**, and **SOCKS** exploitation attempts. Block or alert on:
2. Unauthorized inbound or outbound connections to ports **502 (Modbus)**, **161/162 (SNMP)**, **8291 (MikroTik WinBox)**, or **1080 (SOCKS)**.
3. Excessive DNS requests or email relaying behaviours indicative of botnets.
4. **Flow monitoring** – Deploy NetFlow/sFlow to monitor traffic volumes and destinations. Unusual spikes in outbound SMTP or proxies may indicate compromise.
5. **Log collection** – Centralize logs (router logs, firewall logs, endpoint logs) in a syslog server. Use correlation tools (ELK stack, Wazuh) to detect anomalies.
6. **Anomaly detection with AI** – Integrate open-source ML-based anomaly detectors (e.g., Zeek + ML modules). Use unsupervised models to learn normal traffic patterns and detect deviations. This helps counter AI-assisted reconnaissance by adversaries.

4.4 Physical and acoustic security

1. **RF shielding and cable management** – Use shielded Ethernet cables (CAT6A F/UTP or S/FTP) and route them away from power lines to reduce electromagnetic coupling. Terminate shielding at a single ground point to avoid ground loops.
2. **Soundproofing** – Use heavy curtains, acoustic foam or soundproof panels on windows and walls facing public areas. This attenuates ultrasonic signals that could carry command injections.
3. **Sensor placement** – Avoid placing voice assistants or smart speakers near windows or walls; mount them in the interior of the apartment to reduce exposure to outside ultrasound transmissions.

5 Implementation guidelines

The following table summarises recommended technologies and their purposes. A more detailed implementation plan would be tailored to specific equipment.

Component	Technology / Action	Purpose
Core firewall/router	pfSense/OPNsense on mini-PC	Provides VLANs, advanced firewall, IDS/IPS; regularly patchable
Wi-Fi access points	Ubiquiti Unifi or TP-Link Omada APs	Separate SSIDs; support WPA3; manageable via controller
VLAN segmentation	Trusted, IoT, Guest	Isolates devices; prevents lateral movement

Component	Technology / Action	Purpose
DNS security	DNS over HTTPS/TLS; strict SPF, DKIM, DMARC	Prevents spoofing and DNS hijacking ⁵
Router patching	Firmware updates; disable WinBox, Telnet, API; change defaults	Mitigates RCE vulnerabilities like CVE-2025-10948 ² ; removes unnecessary services
Endpoint security	EDR/antivirus; OS updates; strong passwords	Detects malware; reduces local exploitation
IoT hardening	Firmware updates; disable UPnP; strong Wi-Fi keys; microphone muting	Reduces attack surface; prevents eavesdropping
IDS/IPS	Suricata/Snort; NetFlow monitoring	Detects exploitation of Modbus, SNMP, SOCKS and malspam proxies
Acoustic defense	Physical mute switches; ultrasonic filters; soundproofing	Mitigates inaudible command injection ⁴

6 Operational practices

1. **Patching and updates** – Schedule monthly reviews to apply firmware updates to routers, switches, IoT devices and endpoints. Subscribe to security advisories from vendors (e.g., MikroTik, pfSense). For MikroTik, monitor CVE announcements and update to patched RouterOS releases quickly ² .
2. **Account hygiene** – Use a password manager to generate unique, complex passwords. Rotate passwords periodically. Enable multi-factor authentication where possible.
3. **Email and web hygiene** – Use reputable email providers with spam filtering. Educate household members about phishing emails and suspicious links. Do not reuse passwords across services.
4. **Incident response** – Prepare a simple playbook:
 5. Identify suspicious network behaviour (IDS alerts, unusual traffic).
 6. Isolate the affected device by unplugging or VLAN quarantining.
 7. Reset credentials and reflash firmware if required.
 8. Review logs to understand the attack vector and update firewall/IDS rules accordingly.
9. **Privacy considerations** – Evaluate the necessity of each IoT device. Disable features that are not needed (e.g., remote voice purchasing). Review vendor privacy policies. Use local processing options when available.

7 Conclusion

A residential network is not immune to the sophisticated techniques discussed in industrial training videos or used by botnets and researchers. Misconfigured routers, default passwords, and unsegmented networks can make an apartment a launchpad for larger attacks. By adopting a **defence-in-depth strategy**—hardened infrastructure, network segmentation, endpoint security, monitoring, physical and acoustic safeguards, and disciplined operational practices—a home user can greatly reduce the risk of compromise. This design provides a blueprint for implementing such defences, with controls tailored to counter the specific vectors highlighted in the research and training materials.

1 5 6 13,000 MikroTik Routers Hijacked by Botnet for Malspam and Cyberattacks

<https://thehackernews.com/2025/01/13000-mikrotik-routers-hijacked-by.html>

2 NVD - CVE-2025-10948

<https://nvd.nist.gov/vuln/detail/CVE-2025-10948>

3 4 SUAD: Solid-Channel Ultrasound Injection Attack and Defense to Voice Assistants

<https://arxiv.org/html/2508.02116v1>